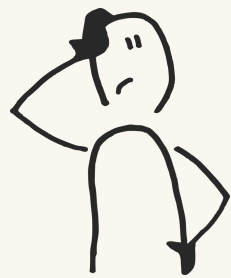


Smart ChatBot for Senior Residential Assistant

Joseph Perez
Hyber Gamboa

INTRODUCTION

Project Overview



Problem

Senior Resident Assistants are often responsible for creating and managing the on-call schedule for their staff. However, scheduling conflicts can frequently arise, making it challenging to adjust shifts and ensure proper coverage.



Solution

We have designed a smart chatbot to assist Senior Resident Assistants by automating the shift-switching process.



Why It Matters

By developing this chatbot, Senior Resident Assistants can more efficiently manage on-call schedules and handle shift changes that occur throughout the year.

INTRODUCTION

Project Objective

Main Goal

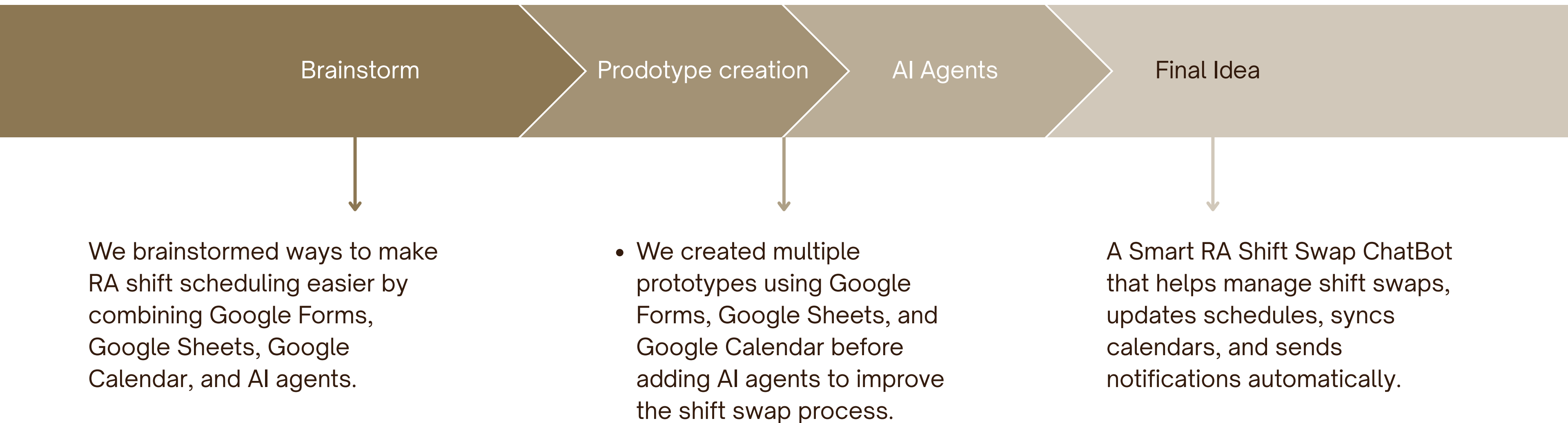
Develop a Smart ChatBot that helps Senior Resident Assistants manage RA on-call schedules by automating shift swaps, updating Google Sheets and Google Calendar, reducing conflicts, and keeping accurate records.

Supporting Objectives

- Collect shift swap requests
- Verify RA availability and rules
- Update Google Sheets automatically
- Sync changes with Google Calendar
- Log all shift changes
- Notify RAs and supervisors



Who We Designed For



Prototype 1

- Created a Google Form for RA shift swap requests
- Responses were saved into Google Sheets
- Helped organize swap details in one place
- Senior RA still had to review and update the schedule manually

Sheets

Shift Swap Request

This is a prototype 1 form for my EGI first

PROTOTYPES

Prototype 2

- Added Google Calendar to the shift swap system
- Swap requests were still collected through Google Forms
- Google Sheets stored and organized the request data
- Calendar events helped RAs see upcoming on-call shifts
- Made the system more useful, but updates still needed review before approval

Sheets

Forms



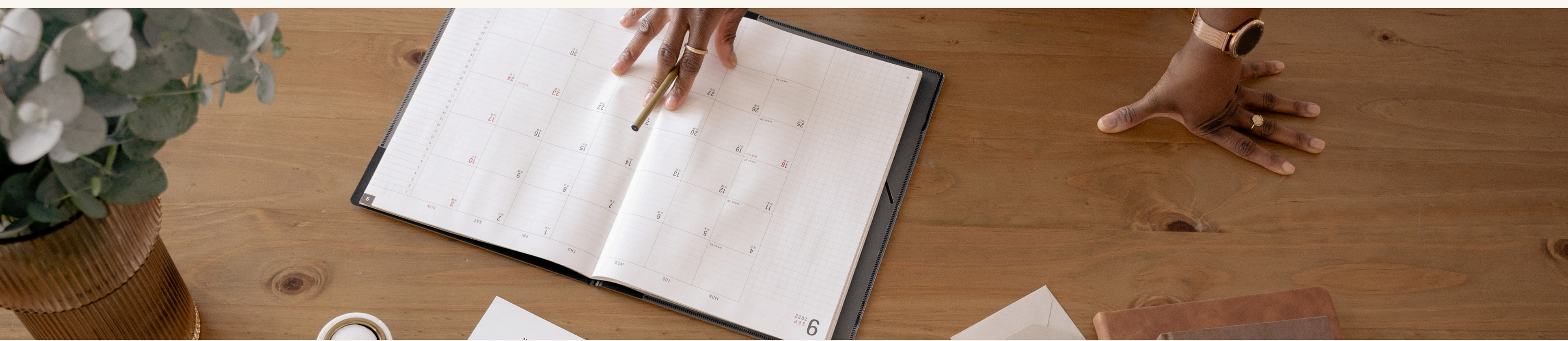


Prototype 3

- Added an AI chatbot to help manage shift swaps
- Chatbot understands RA requests in normal language
- Agents check rules, update the schedule, sync the calendar, and log changes
- Helps reduce manual work for Senior RAs
- Makes the shift swap process faster and more organized

Sheets

Forms



Prototype 4

- Combined Google Forms, Sheets, Calendar, and AI agents
- Added a Streamlit page for easier user interaction
- Agents help validate swaps, update schedules, sync calendars, and send notifications
- Includes guardrails to keep the chatbot focused on RA scheduling
- Final version makes shift management more automated and organized

Forms

Sheets

AI Agents

1

The AI agents work together to manage the RA shift swap process.

2

The Intake Agent understands the request and collects important details.

3

The Policy Agent checks if the swap follows the rules before any changes are made.

4

The other agents update the schedule, sync the calendar, log the change, and send notifications.

CHALLENGES FACED

App

The Streamlit app gives users a simple web page to submit shift swap requests and interact with the chatbot. It organizes the schedule, swap details, and chatbot responses in one place so Senior RAs can manage changes more easily.

This makes the system easier to test, present, and use compared to only working through Google Forms and spreadsheets.

What We Learned

Personal Skills Gained

We improved our skills in using Google Forms, Google Sheets, Google Calendar, Streamlit, and AI agents to build a working system.

Problem-Solving Insights

We learned that scheduling problems need clear rules, organized data, and a reliable way to track every change.

Team Collaboration Lessons

We learned how to divide tasks, combine ideas, and communicate clearly to make each part of the project work together.

Adaptability

We learned how to adjust our project as new challenges came up, such as improving prototypes, adding automation, and making the system easier to use.

The End

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